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Studierichting: Informatica
Oefeningenreeks 5K nr. 1.1

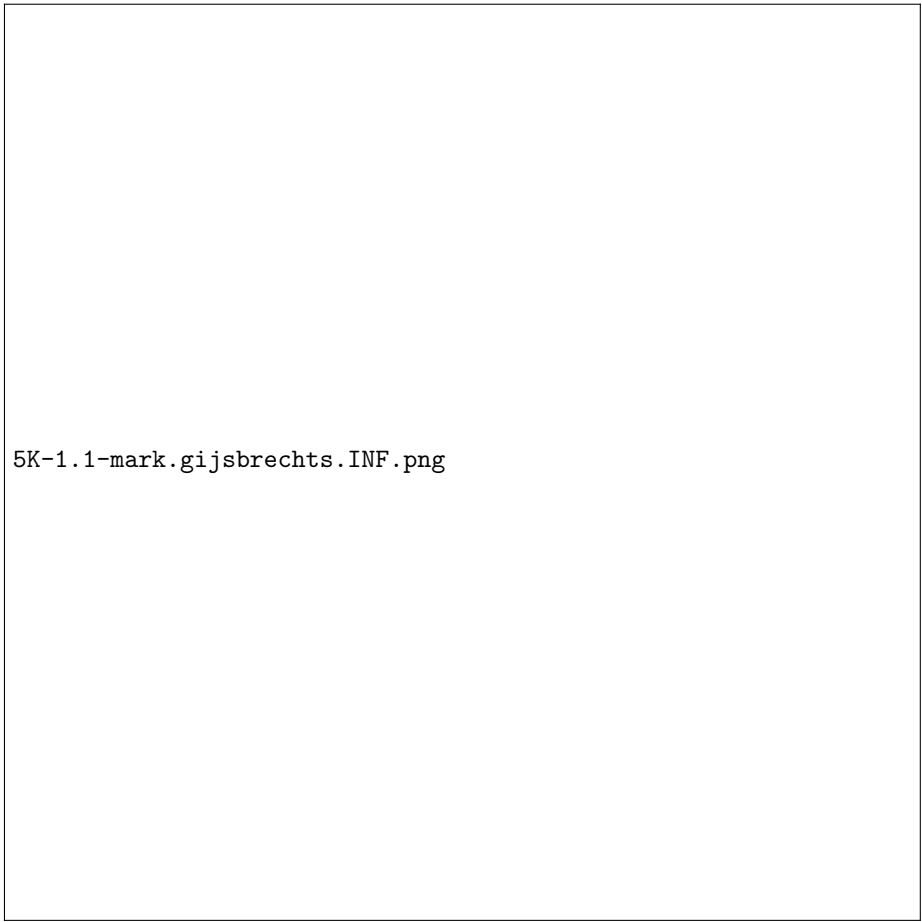
$$r = 2 \cos \theta$$

$$r^2 = 4 \cos^2 \theta$$

$$r' = -2 \sin \theta \Rightarrow (r')^2 = 4 \sin^2 \theta$$

$$\begin{aligned} L &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4 \cos^2 \theta + 4 \sin^2 \theta} d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4 (\cos^2 \theta + \sin^2 \theta)} d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4} d\theta \\ &= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \\ &= 2 [\theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= 2 \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] \\ &= 2\pi \end{aligned}$$

References



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